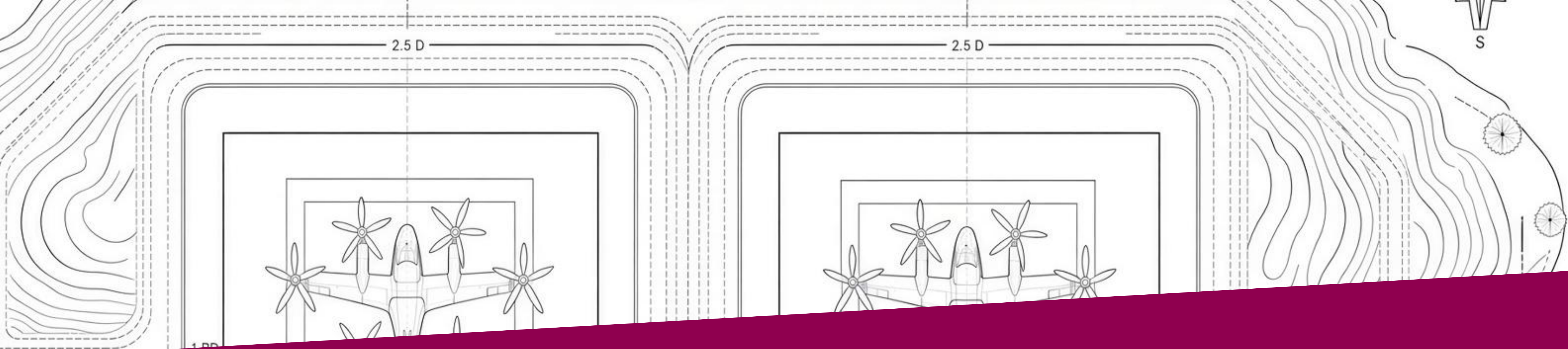




Accelerating MATSim-Based Vertiport Siting via Facility Location Surrogate Space Mapping

Ziyue Song | MUM 2026 | 28 September 2026

MATSim
Multi-Agent Transport Simulation

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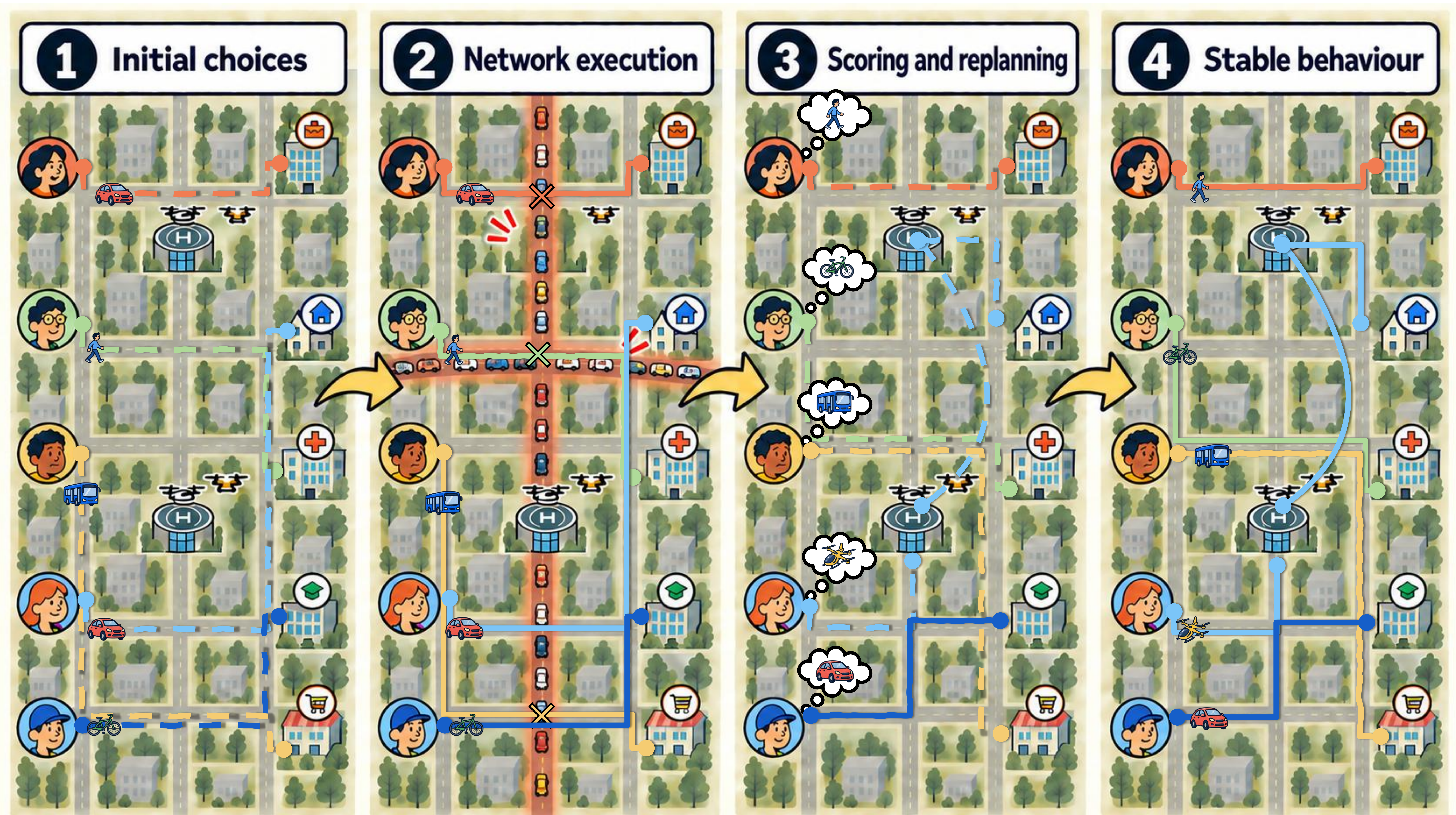
Reporter:

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Evaluating vertiport siting requires behavioural simulation

MATSim Resolves Traveller-network Feedback Through Iteration



High-fidelity Evaluation Creates an Optimisation Bottleneck

Vertiport siting problem: $\min_X \{-Share_{UAM}(X), B(X)\}$

X : open/close each site (binary)
 $Share_{UAM}(X)$: UAM mode share
 $B(X)$: vertiport numbers

Calculational Burden

HIGH



LOW

Behavioural Fidelity

HIGH

● Clustering / GIS

Captures: demand hotspots and spatial coverage
Limitation: no traveller choice or network feedback

● Static FLP / IP / MILP

Captures: discrete sites, capacity, cost and coverage
Limitation: fixed demand and simplified travel impedance

● Data-driven surrogate

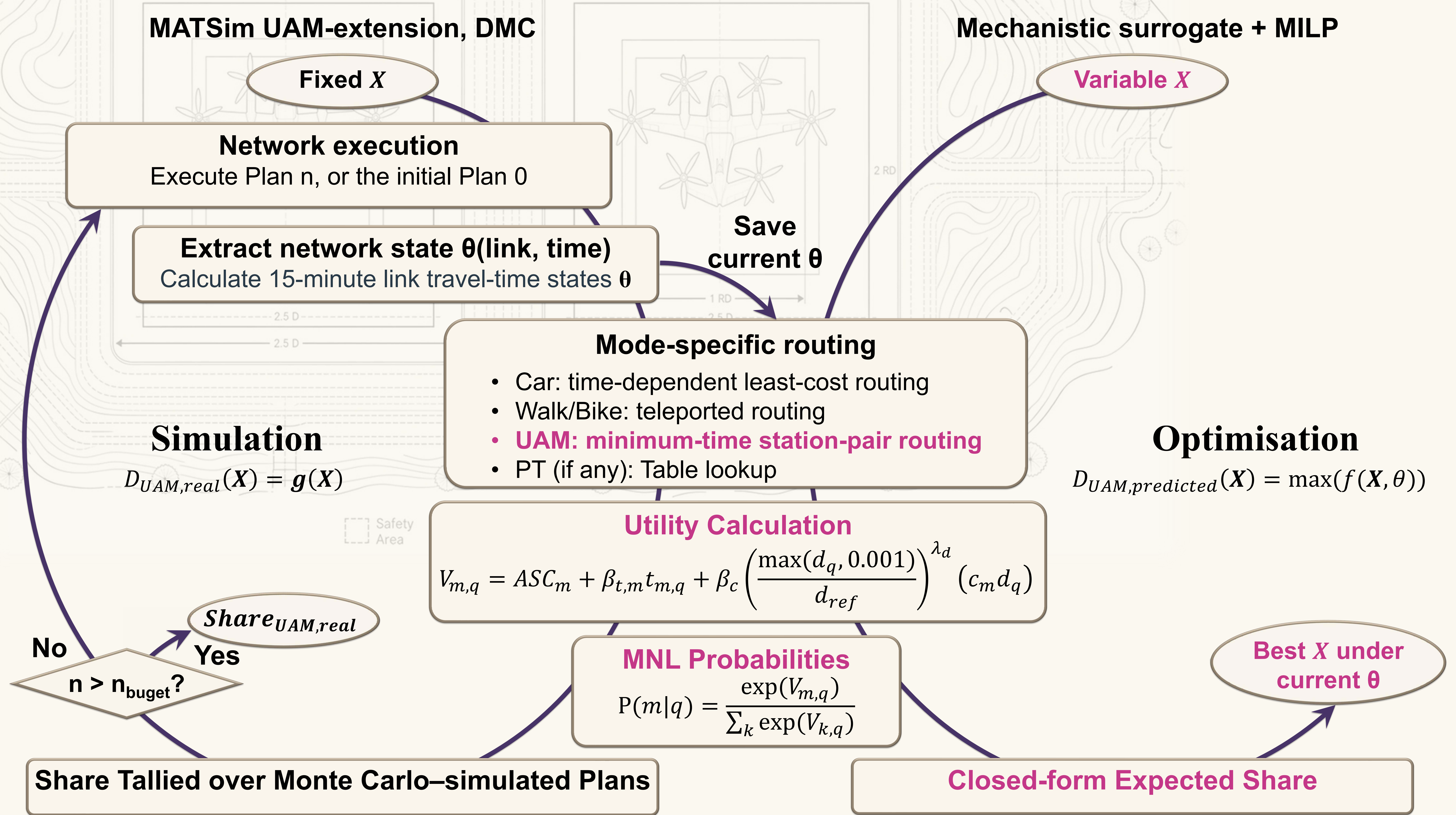
Captures: rapid screening of many candidate layouts
Limitation: depends on training data and weakens outside its sample

● GA / NSGA-II + MATSim

Captures: heterogeneous agents and multimodal network feedback
Limitation: very high runtime and noisy convergence

○ This study

A Mechanistic Surrogate for MATSim-UAM

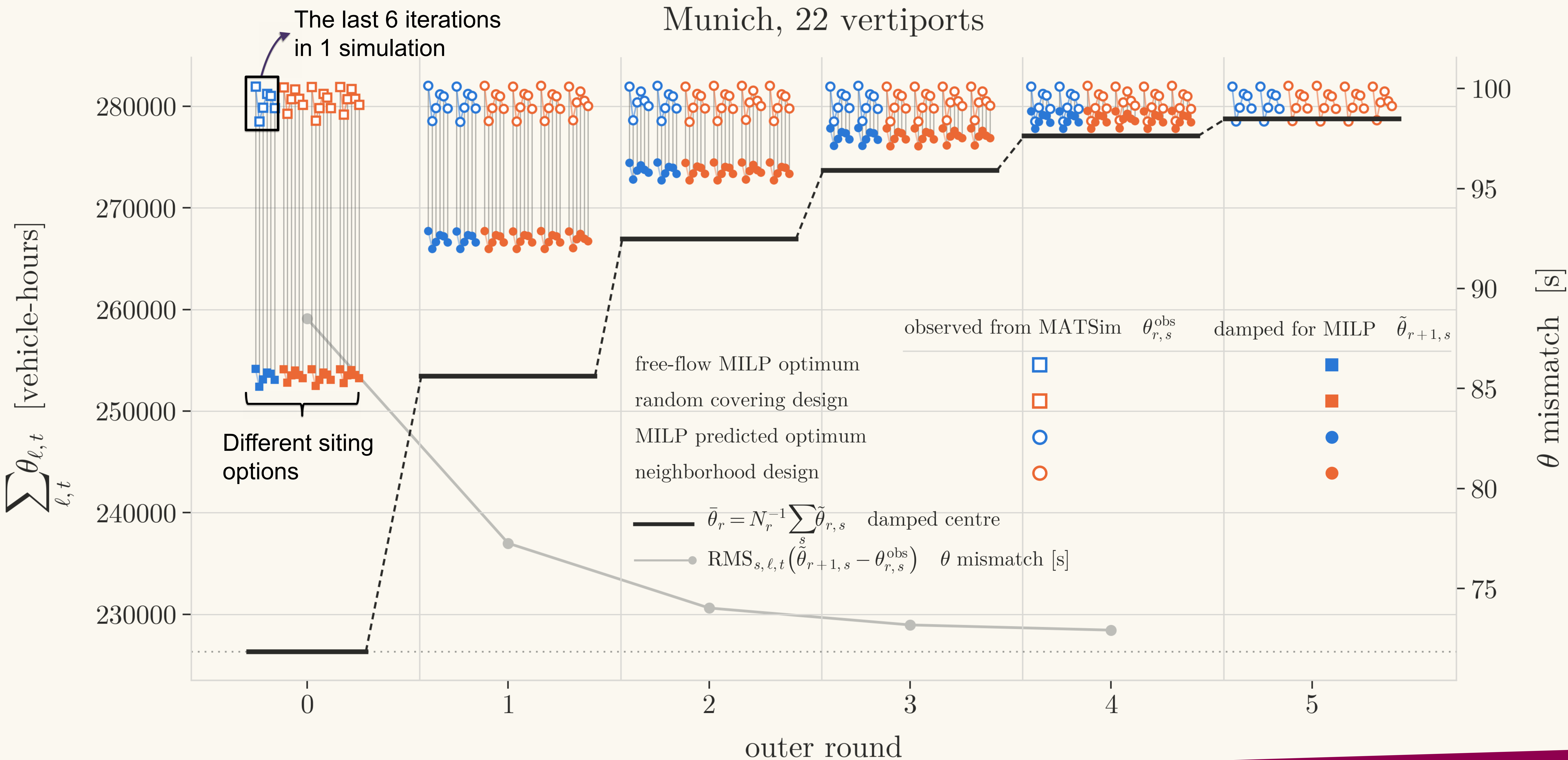


Round 0

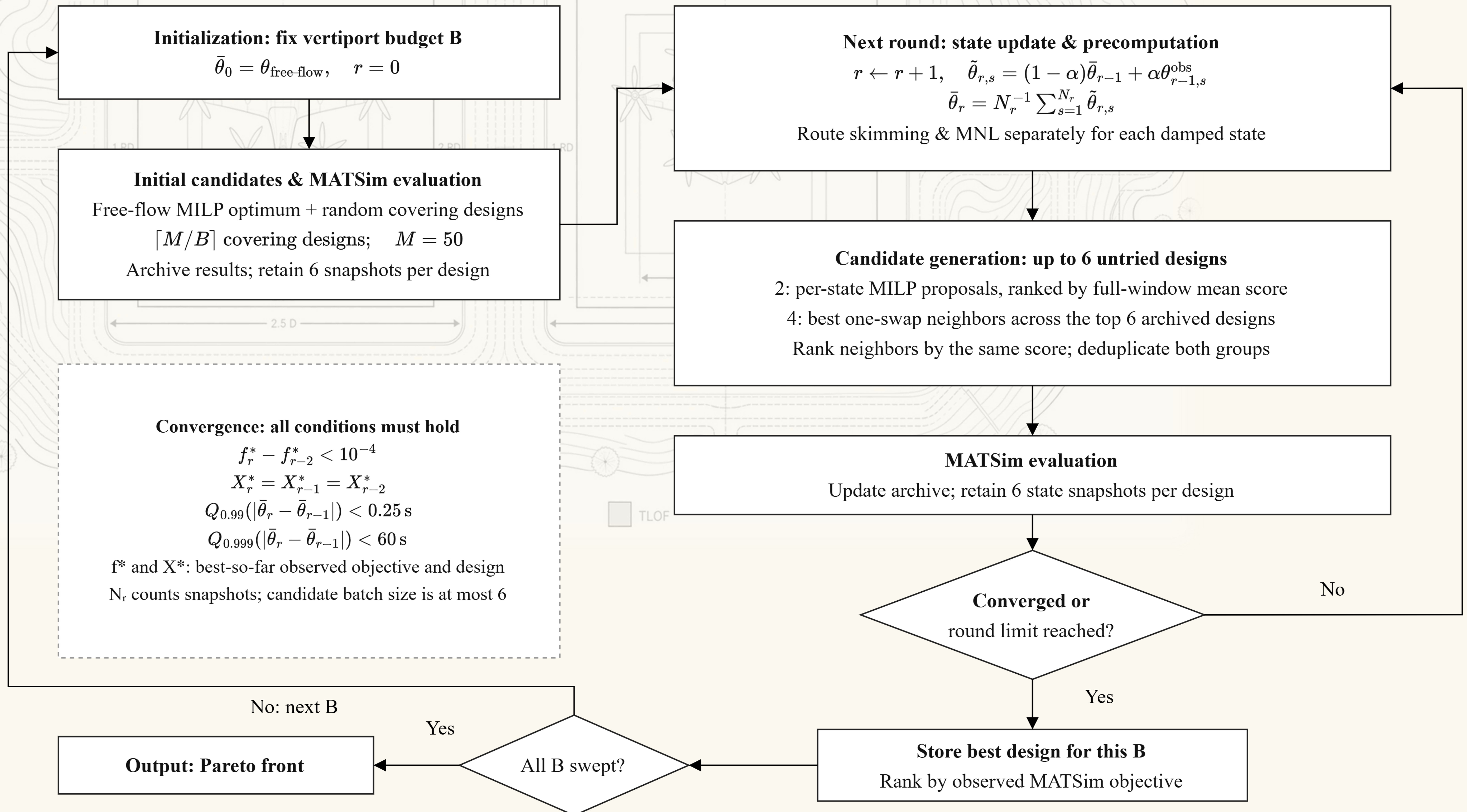
- 1: free-flow MILP optimum
- N/B: random covering designs

Round 1~n

- 2: per-state MILP proposals, ranked by full-window mean score
- 4: best one-swap neighbors across the top 6 archived designs



Space Mapping Optimisation Framework



Corsica, FR

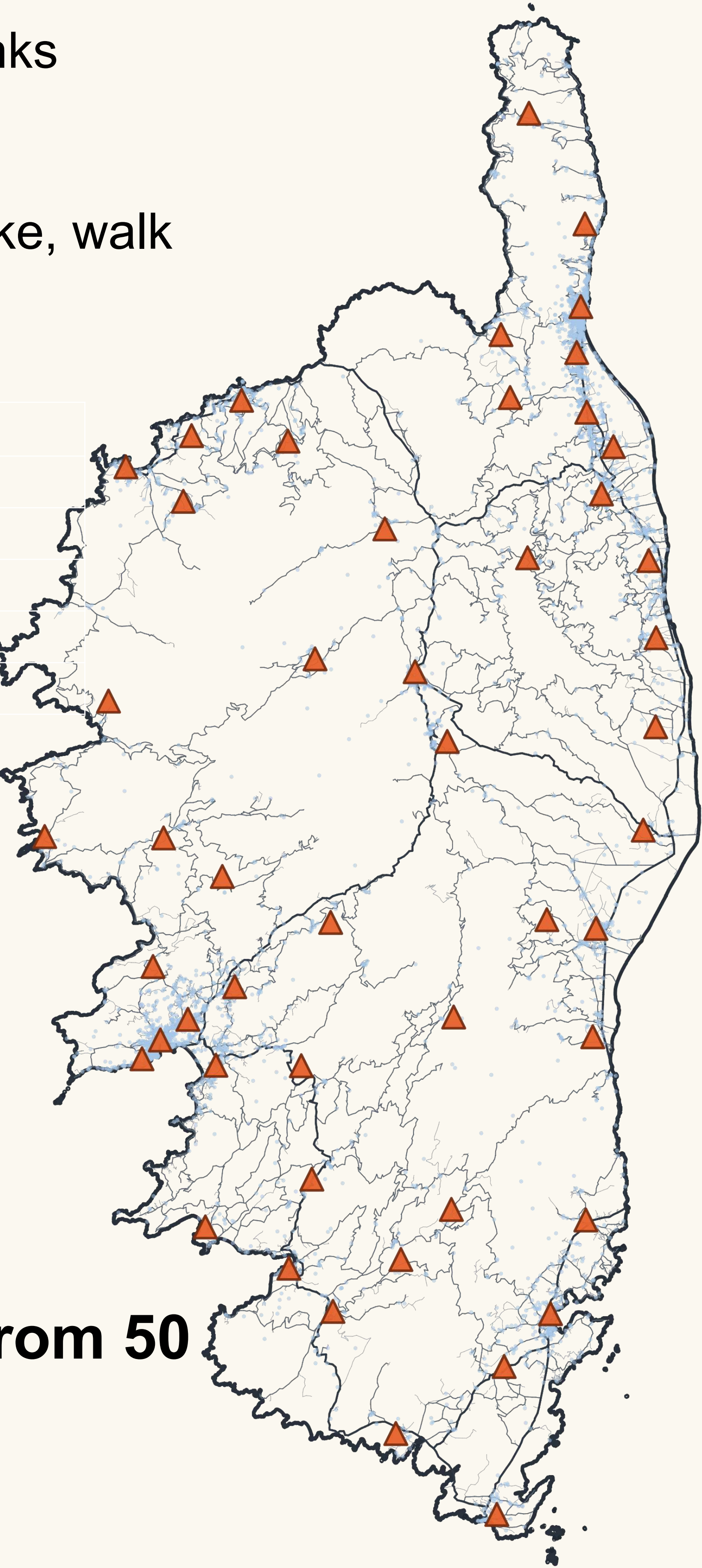
Population: 3,162 agents, 1% sample
Area: 8,752 km²
Network: 34k nodes / 70k links
Vertiports: 50 candidates
OD pairs: 7,764 pairs
Modes: car, car_passenger, bike, walk

Utility ref: Hörl & Balac 2021

Mode	ASC_m	$\beta_{t,m}(/min)$	$c_m(€/km)$	VOT
car	1.35	-0.060	0.15	17.5
walk	1.43	-0.150	0	43.7
bike	-2.00	-0.050	0	14.6
uam	-1.28	-0.153	1.26	44.6
car_pass	captive, ±700			

Select B vertiports from 50

$$C_{50}^B = \frac{50!}{B!(50 - B)!}$$

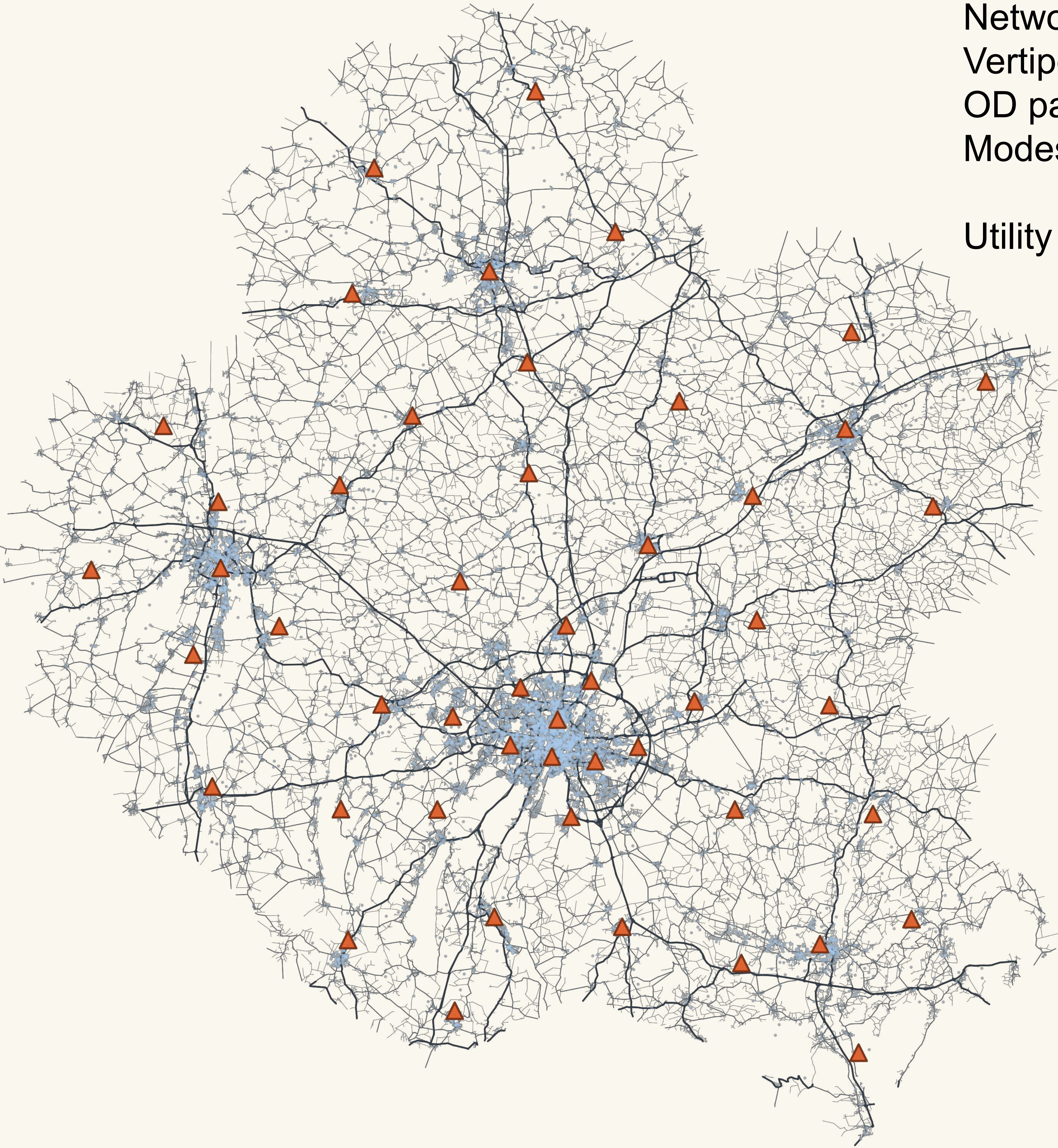


Munich, DE

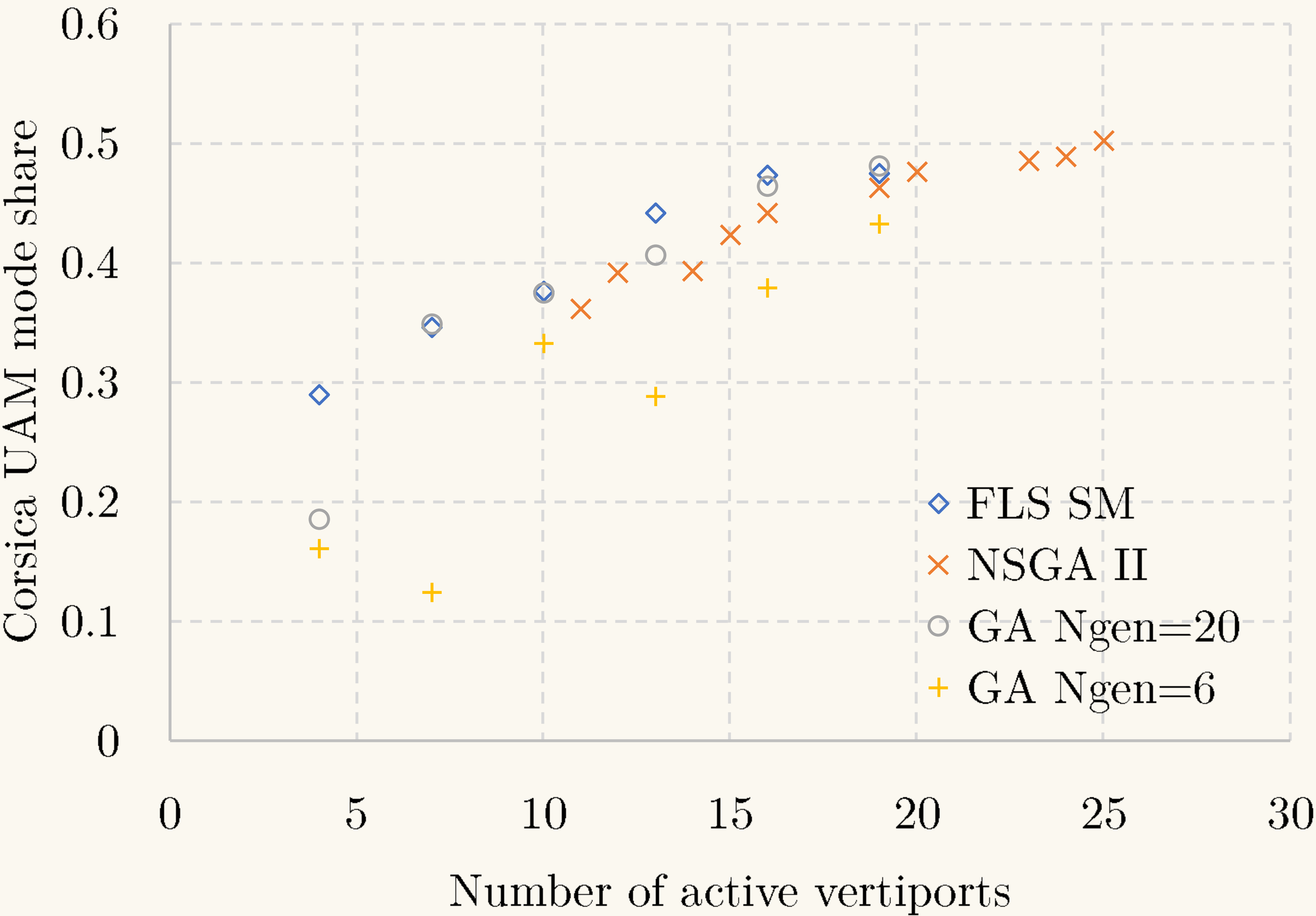
Population: 3,240 agents, 0.05% sample
Area: 14,882 km²
Network: 213k nodes / 499k links
Vertiports: 50 candidates
OD pairs: 5,349 pairs
Modes: car, car_passenger, bike, walk, PT

Utility ref: Fu et al. (2019)

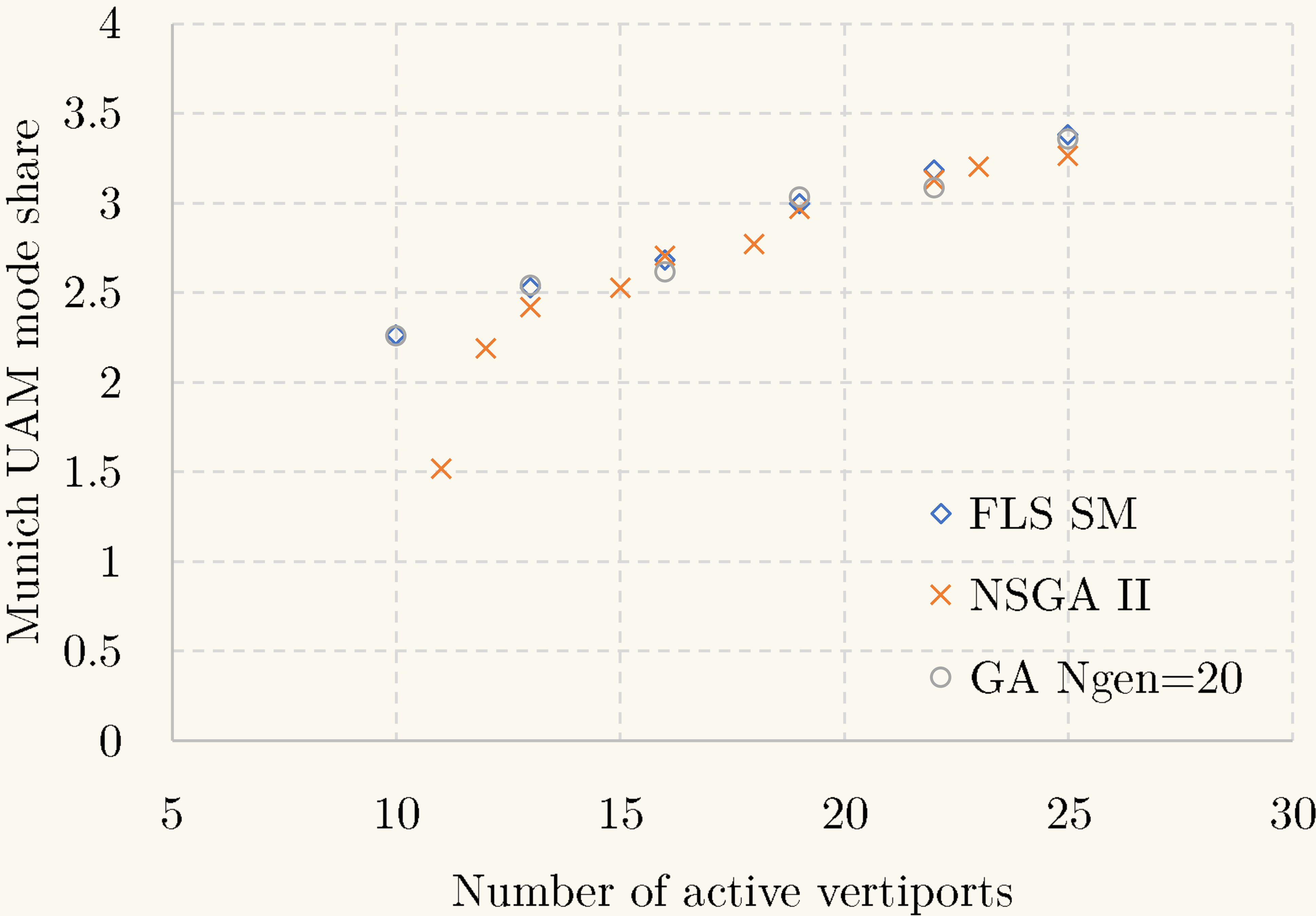
Mode	ASC_m	$\beta_{t,m}(/min)$	$c_m(€/km)$	VOT
car	0.00	-0.0946	0.15	27.6
walk	0.00	$\beta^{iv}t, m = -0.09$ $\beta^{ww}t, m = -0.11$	0	27.5/ 31.1
bike	1.43	-0.150	0	43.7
uam	-2.00	-0.050	0	14.6
car_pass	captive, ±700			



— Road network • Trip ends (OD endpoints) ▲ Candidate vertiports



Pareto frontier of Corsica



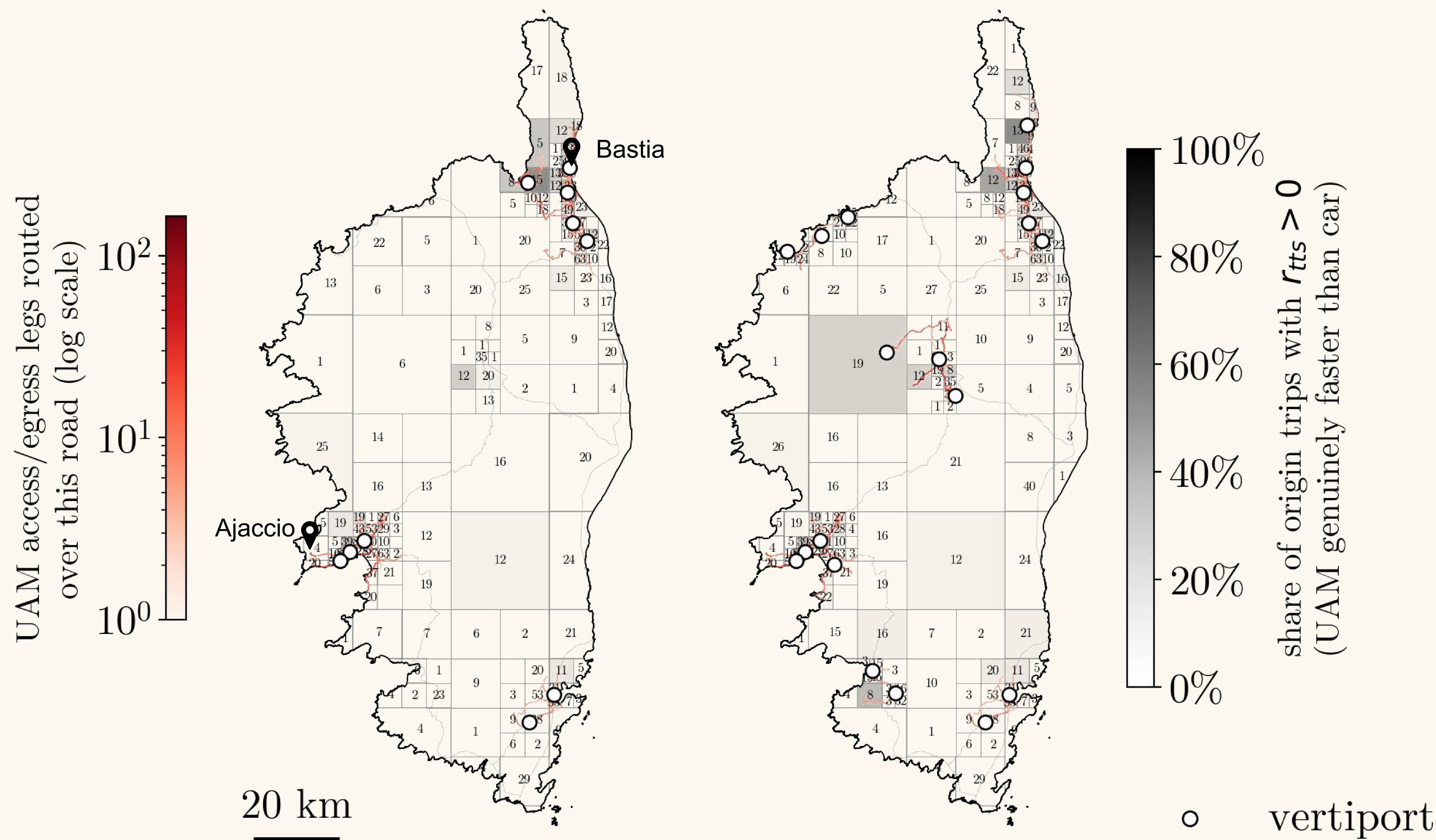
Pareto frontier of Munich

Comparison of optimisation methods across scenarios

Method	Corsica				Munich			
	HV	Evals	Evals/pt.	Time(h)	HV	Evals	Evals/pt.	Time(h)
FLS-SM	0.759	211	35.2	17.5	0.748	195	32.5	53.0
GA-Ngen20	0.707	1200	200	47.5	0.739	1200	200	127.0

HV: hypervolume, normalised area a front covers toward the ideal corner
Evals: the number of true MATSim runs a method spent to produce its whole front

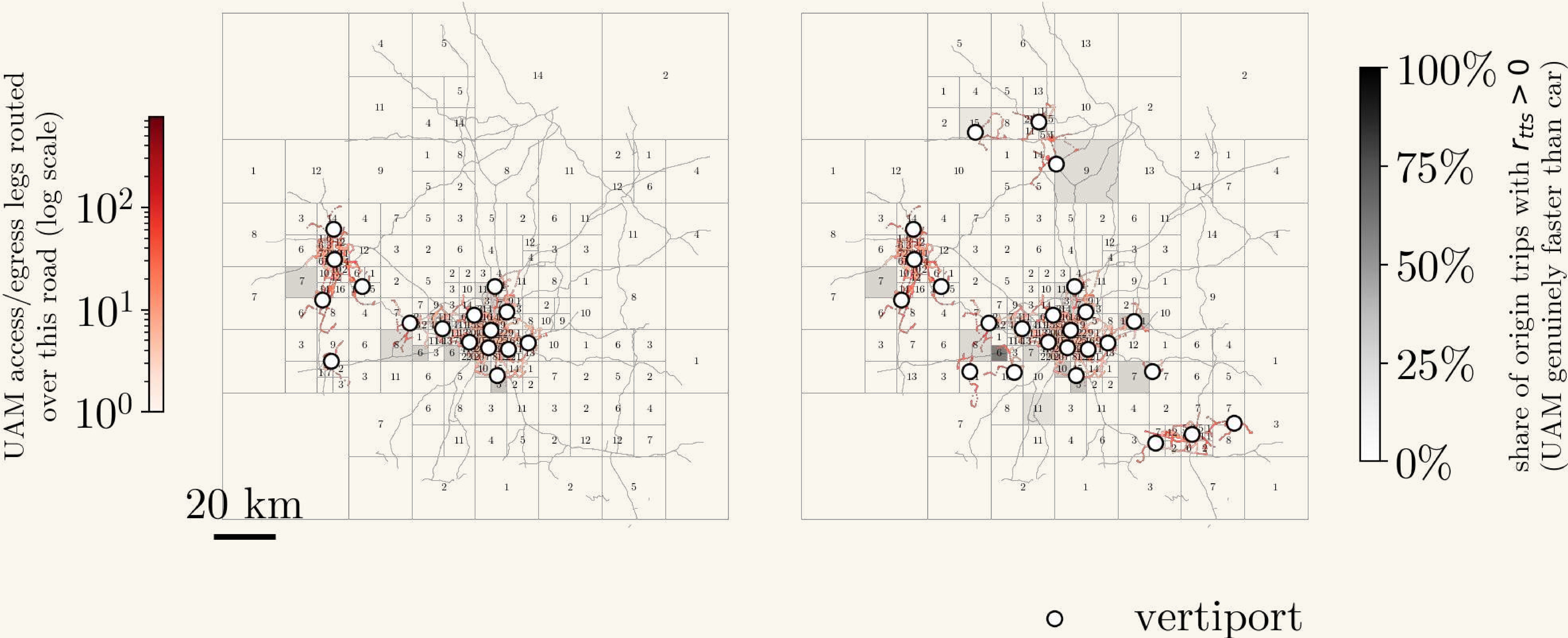
Share of trips with positive r_{tts}
B10 (10 vertiports) B19 (19 vertiports)

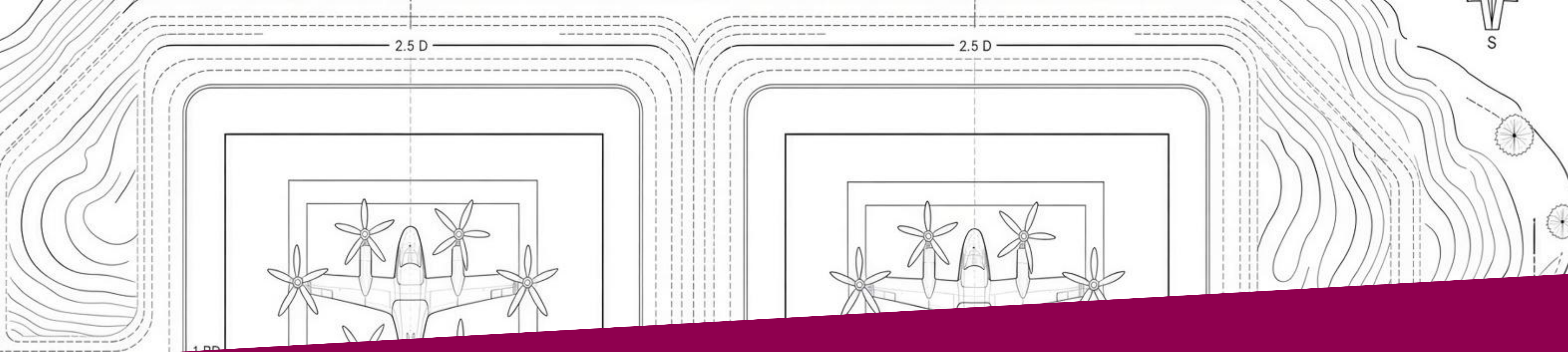


Share of trips with positive r_{tts}

B16 (16 vertiports)

B25 (25 vertiports)





Thank you for your attention!

Conclusions:

- FLS-SM combines a mechanistic surrogate with MATSim.
- Tests in Corsica and Munich demonstrate its broad applicability.
- FLS-SM uses 82–84% fewer evaluations than GA with higher HV.

Future Work:

- Tune hyperparameters and damping strategies to improve performance.
- Test generalisability to other tasks, such as road-policy optimisation.
- Extend beyond DMC to alternative mode-choice approaches, such as SMC.

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